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TIME VARYING PERFORMANCE INDICATION SYSTEM FOR CONNECTED EQUIPMENT

Abstract

A system includes equipment operable to affect, monitor, or control one or more variable states or conditions in a building, a sensor configured to measure a variable relating to performance of the equipment, and a control system. The control system is configured to perform performance checks for the equipment using a measurement from the sensor and determine individual performance check indicators based on the performance checks. The individual performance check indicators indicate events having different timings and severities. The control system is configured to generate an overall performance index for the equipment using the individual performance check indicators and weights based on the different timings and severities of the events indicated by the individual performance check indicators. The control system is configured to cause an adjustment to the equipment by initiating or executing a preventative maintenance action for the equipment based on the overall performance index.

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Background/Summary

CROSS-REFERENCE TO RELATED PATENT APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 18/105,721 filed Feb. 3, 2023, which is a continuation of U.S. patent application Ser. No. 17/400,950 filed Aug. 12, 2021 (now U.S. Pat. No. 11,630,453), which is a continuation of U.S. patent application Ser. No. 16/245,122, filed Jan. 10, 2019 (now U.S. Pat. No. 11,092,954), all of which are incorporated by reference herein in their entireties.

BACKGROUND

[0002] The present disclosure relates generally to building management systems. The present disclosure relates more particularly to fault detection and generating an overall performance index for connected equipment in a building management system. A building management system (BMS) is, in general, a system of devices configured to control, monitor, and manage equipment in or around a building or building area. A BMS can include, for example, a HVAC system, a security system, a lighting system, a fire alerting system, any other system that is capable of managing building functions or devices, or any combination thereof.

[0003] Systems and devices in a BMS often generate temporal or time-series data that can be analyzed to determine the performance of the BMS and the various components thereof. The data generated by the BMS can include measured or calculated values that exhibit statistical characteristics and provide information about how the corresponding system or process (e.g., a temperature control process, a flow control process, etc.) is performing in terms of error from its setpoint. These data can be examined and alert a user to repair the fault before it becomes more severe when the monitored system or process begins to degrade in performance.

SUMMARY

[0004] One implementation of the present disclosure is a system for generating a performance index for connected equipment. The system includes the connected equipment and a time varying performance indication system. The connected equipment can be configured to measure a plurality of monitored variables and generate a plurality of status codes. The time varying performance indication system can be configured to obtain data points

of the plurality of monitored variables and the plurality of status codes from past N time units, and N is a number. The time varying performance indication system can be configured to obtain a plurality of connected equipment specific parameters that are parameters specific to the connected equipment. The time varying performance indication system can be configured to perform a plurality of first performance checks for the connected equipment using the plurality of status codes from the past N time units. The time varying performance indication system can be configured to perform a plurality of second performance checks for the connected equipment using the data points of the plurality of monitored variables from the past N time units, the plurality of connected equipment specific parameters, and a plurality of predetermined rules. The time varying performance indication system can be configured to determine a plurality of individual performance check indicators based on the first performance checks and the second performance checks using a plurality of first weights each determined based on a different timing. The time varying performance indication system can be configured to generate an overall performance index for the connected equipment using the plurality of individual performance check indicators and a plurality of second weights.

[0005] In some embodiments, the connected equipment is a chiller.

[0006] In some embodiments, the plurality of second performance checks are performed by applying the data points of the plurality of monitored variables from the past N time units and the plurality of connected equipment specific parameters to the plurality of predetermined rules.

[0007] In some embodiments, each of the plurality of second weights is determined based on a predetermined degree of severity of a respective first performance check or a respective second performance check.

[0008] In some embodiments, the time varying performance indication system can be configured to determine that a total runtime of the connected equipment in a past time window, and generate the overall performance index only when the total runtime of the connected equipment in the past time window satisfies a predetermined threshold.

[0009] In some embodiments, the time varying performance indication system can be configured to cause an adjustment to the connected equipment based on the overall performance index generated for the connected equipment.

[0010] In some embodiments, the time units are days, hours, minutes, seconds, weeks, months, or years.

[0011] In some embodiments, the first performance checks are status checks and the second performance checks are health checks.

[0012] Another implementation of the present disclosure is a method for generating a performance index for connected equipment. The method includes obtaining, by a time varying performance indication system comprising a processing circuit and memory, data points of a plurality of monitored variables and a plurality of status codes from past N time units. N is a number. The plurality of monitored variables are measured by connected equipment, and the plurality of status codes are generated by the connected equipment. The method includes obtaining, by the time varying performance indication system, a plurality of connected equipment specific parameters that are parameters specific to the connected equipment. The method includes performing, by the time varying performance indication system, a plurality of first performance checks for the connected equipment using the plurality of status codes from the past N time units. The method includes performing, by the time varying performance indication system, a plurality of second performance checks for the connected equipment using the data points of the plurality of monitored variables from the past N time units, the plurality of connected equipment specific parameters, and a plurality of predetermined rules. The method includes determining, by the time varying performance indication system, a plurality of individual performance check indicators based on the first performance checks and the second performance checks using a plurality of first weights each determined based on a different timing. The method includes generating, by the time varying performance indication system, an overall performance index for the connected equipment using the plurality of individual performance check indicators and a plurality of second weights.

[0013] In some embodiments, the method includes determining that a total runtime of the connect equipment in a past time window, and generating the overall performance index only when the total runtime of the connect equipment in the past time window satisfies a predetermined threshold.

[0014] In some embodiments, the method includes causing an adjustment to the connect equipment based on the overall performance index generated for the connected equipment.

[0015] Another implementation of the present disclosure is a non-transitory computer-readable medium. The non-transitory computer-readable medium stores computer-executable instructions that when executed by at least one processor, causing the at least one processor to perform operations for generating a performance index for connected equipment. The operations include obtaining data points of a plurality of monitored variables and a plurality of status codes from past N time units. N is a number. The plurality of monitored variables are measured by connected equipment, and the plurality of status codes are generated by the connected equipment. The operations include obtaining a plurality of connected equipment specific parameters that are parameters specific to the connected equipment. The operations include performing a plurality of first performance checks for the connected equipment using the plurality of status codes from the past N time units. The operations include performing a plurality of second performance checks for the connected equipment using the data points of the plurality of monitored variables from the past N time units, the plurality of connected equipment specific parameters, and a plurality of predetermined rules. The operations include determining a plurality of individual performance check indicators based on the first performance checks and the second performance checks using a plurality of first weights each determined based on a different timing. The operations include generating an overall performance index for the connected equipment using the plurality of individual performance check indicators and a plurality of second weights.

[0016] In some embodiments, the operations include determining that a total runtime of the connect equipment in a past time window, and generating the overall performance index only when the total runtime of the connect equipment in the past time window satisfies a predetermined threshold.

[0017] In some embodiments, the operations include causing an adjustment to the connect equipment based on the overall performance index generated for the connected equipment.

[0018] Those skilled in the art will appreciate that the summary is illustrative only and is not intended to be in any way limiting. Other aspects, inventive features, and advantages of the devices and/or processes described herein, as defined solely by the claims, will become apparent in the detailed description set forth herein and taken in conjunction with the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0019] FIG. 1 is a drawing of a building equipped with a HVAC system, according to some embodiments.

[0020] FIG. 2 is a schematic diagram of a waterside system which can be used in conjunction with the building of FIG. 1, according to some embodiments.

[0021] FIG. 3 is a schematic diagram of an airside system which can be used in conjunction with the building of FIG. 1, according to some embodiments.

[0022] FIG. 4 is a block diagram of a building management system (BMS) which can be used to monitor and control the building of FIG. 1, according to some embodiments.

[0023] FIG. 5 is a block diagram of another BMS which can be used to monitor and control the building of FIG. 1 and includes a time varying performance indication system, according to some embodiments.

[0024] FIG. 6 is a block diagram of another BMS including the time varying performance indication system for generating a performance index for connected equipment, according to some embodiments.

[0025] FIG. 7 is a schematic diagram of a chiller, which is an example of a type of connected equipment which can report monitored variables and status information to the time varying performance indication system, according to some embodiments.

[0026] FIG. **8** is a block diagram of a time varying performance indication system of generating a performance index for connected equipment, according to some embodiments.

[0027] FIG. **9** is a high level flow diagram illustrating a process of generating a performance index for connected equipment, according to some embodiments.

[0028] FIG. **10** is a flow diagram illustrating a process of generating a performance index for connected equipment, according to some embodiments.

[0029] FIG. **11** is another flow diagram illustrating a process of generating a performance index for connected equipment, according to some embodiments.

[0030] FIG. **12** is another flow diagram illustrating a process of generating a performance index for connected equipment, according to some embodiments.

[0031] FIG. **13** is a graph illustrating the exponential decay function with a tau value of two, according to some embodiments.

[0032] FIG. **14** is an example scenario of performance checks for generating an overall performance index, according to some embodiments.

[0033] FIG. **15** is an example user interface illustrating an example of the calculated performance index over time, according to some embodiments.

[0034] FIG. **16** is another flow diagram illustrating a process of generating a performance index for connected equipment, according to some embodiments.

DETAILED DESCRIPTION

[0035] Following below are more detailed descriptions of various concepts related to, and implementations of systems, methods, and apparatuses for generating time varying performance indications for connected equipment in a building management system. Before turning to the more detailed descriptions and figures, which illustrate the exemplary embodiments in detail, it should be understood that the application is not limited to the details or methodology set forth in the descriptions or illustrated in the figures. It should also be understood that the terminology is for the purpose of description only and should not be regarded as limiting in any way.

Building HVAC Systems and Building Management Systems

[0036] Referring now to FIGS. **1-5**, several building management systems (BMS) and HVAC systems in which the systems and methods of the present disclosure can be implemented are shown, according to some embodiments. In brief overview, FIG. **1** shows a building **10** equipped with a HVAC system **100**. FIG. **2** is a block diagram of a waterside system **200** which can be used to serve building **10**. FIG. **3** is a block diagram of an airside system **300** which can be used to serve building **10**. FIG. **4** is a block diagram of a BMS which can be used to monitor and control building **10**. FIG. **5** is a block diagram of another BMS which can be used to monitor and control building **10**.

Building **10** and HVAC System **100**

[0037] Referring particularly to FIG. **1**, a perspective view of building **10** is shown. Building **10** is served by a BMS. A BMS is, in general, a system of devices configured to control, monitor, and manage equipment in or around a building or building area. A BMS can include, for example, a HVAC system, a security system, a lighting system, a fire alerting system, any other system that is capable of managing building functions or devices, or any combination thereof.

[0038] The BMS that serves building **10** includes an HVAC system **100**. HVAC system **100** can include a plurality of HVAC devices (e.g., heaters, chillers, air handling units, pumps, fans, thermal energy storage, etc.) configured to provide heating, cooling, ventilation, or other services for building **10**. For example, HVAC system **100** is shown to include a waterside system **120** and an airside system **130**. Waterside system **120** may provide a heated or chilled fluid to an air handling unit of airside system **130**. Airside system **130** may use the heated or chilled fluid to heat or cool an airflow provided to building **10**. An exemplary waterside system and airside system which can be used in HVAC system **100** are described in greater detail with reference to FIGS. **2** and **3**.

[0039] HVAC system **100** is shown to include a chiller **102**, a boiler **104**, and a rooftop air handling unit (AHU) **106**. Waterside system **120** may use boiler **104** and chiller **102** to heat or cool a working fluid (e.g., water, glycol, etc.) and may circulate the working fluid to AHU **106**. In various embodiments, the HVAC devices of waterside system **120** can be located in or around building **10** (as shown in FIG. **1**) or at an offsite location such as a central plant (e.g., a chiller plant, a steam plant, a heat plant, etc.). The working fluid can be heated in boiler **104** or cooled in chiller **102**, depending on whether heating or cooling is required in building **10**. Boiler **104** may add heat to the circulated fluid, for example, by burning a combustible material (e.g., natural gas) or using an electric heating element. Chiller **102** may place the circulated fluid in a heat exchange relationship with another fluid (e.g., a refrigerant) in a heat exchanger (e.g., an evaporator) to absorb heat from the circulated fluid. The working fluid from chiller **102** and/or boiler **104** can be transported to AHU **106** via piping **108**.

[0040] AHU **106** may place the working fluid in a heat exchange relationship with an airflow passing through AHU **106** (e.g., via one or more stages of cooling coils and/or heating coils). The airflow can be, for example, outside air, return air from within building **10**, or a combination of both. AHU **106** may transfer heat between the airflow and the working fluid to provide heating or cooling for the airflow. For example, AHU **106** can include one or more fans or blowers configured to pass the airflow over or through a heat exchanger containing the working fluid. The working fluid may then return to chiller **102** or boiler **104** via piping **110**.

[0041] Airside system **130** may deliver the airflow supplied by AHU **106** (i.e., the supply airflow) to building **10** via air supply ducts **112** and may provide return air from building **10** to AHU **106** via air return ducts **114**. In some embodiments, airside system **130** includes multiple variable air volume (VAV) units **116**. For example, airside system **130** is shown to include a separate VAV unit **116** on each floor or zone of building **10**. VAV units **116** can include dampers or other flow control elements that can be operated to control an amount of the supply airflow provided to individual zones of building **10**. In other embodiments, airside system **130** delivers the supply airflow into one or more zones of building **10** (e.g., via supply ducts **112**) without using intermediate VAV units **116** or other flow control elements. AHU **106** can include various sensors (e.g., temperature sensors, pressure sensors, etc.) configured to measure attributes of the supply airflow. AHU **106** may receive input from sensors located within AHU **106** and/or within the building zone and may adjust the flow rate, temperature, or other attributes of the supply airflow through AHU **106** to achieve setpoint conditions for the building zone.

Waterside System **200**

[0042] Referring now to FIG. **2**, a block diagram of a waterside system **200** is shown, according to some embodiments. In various embodiments, waterside system **200** may supplement or replace waterside system **120** in HVAC system **100** or can be implemented separate from HVAC system **100**. When implemented in HVAC system **100**, waterside system **200** can include a subset of the HVAC devices in HVAC system **100** (e.g., boiler **104**, chiller **102**, pumps, valves, etc.) and may operate to supply a heated or chilled fluid to AHU **106**. The HVAC devices of waterside system **200** can be located within building **10** (e.g., as components of waterside system **120**) or at an offsite location such as a central plant.

[0043] In FIG. **2**, waterside system **200** is shown as a central plant having a plurality of subplants **202-212**. Subplants **202-212** are shown to include a heater subplant **202**, a heat recovery chiller subplant **204**, a chiller subplant **206**, a cooling tower subplant **208**, a hot thermal energy storage (TES) subplant **210**, and a cold thermal energy storage (TES) subplant **212**. Subplants **202-212** consume resources (e.g., water, natural gas, electricity, etc.) from utilities to serve thermal energy loads (e.g., hot water, cold water, heating, cooling, etc.) of a building or campus. For example, heater subplant **202** can be configured to heat water in a hot water loop **214** that circulates the hot water between heater subplant **202** and building **10**. Chiller subplant **206** can be configured to chill water in a cold water loop **216** that circulates the cold water between chiller subplant **206** and building **10**. Heat recovery chiller subplant **204** can be configured to transfer heat from cold water loop **216** to hot water loop **214** to provide additional heating for the hot water and additional cooling for the cold water. Condenser water loop **218** may absorb heat from the cold water in chiller subplant **206** and reject the absorbed heat in cooling tower subplant **208** or transfer the absorbed heat to hot water loop **214**. Hot TES subplant **210** and cold TES subplant **212** may store hot and cold thermal energy, respectively, for subsequent use.

[0044] Hot water loop **214** and cold water loop **216** may deliver the heated and/or chilled water to air handlers located on the rooftop of building **10** (e.g., AHU **106**) or to individual floors or zones of building **10** (e.g., VAV units **116**). The air handlers push air past heat exchangers (e.g., heating coils or cooling coils) through which the water flows to provide heating or cooling for the air. The heated or cooled air can be delivered to individual zones of building **10** to serve thermal energy loads of building **10**. The water then returns to subplants **202-212** to receive further heating or cooling. [0045] Although subplants **202-212** are shown and described as heating and cooling water for circulation to a building, it is understood that any other type of working fluid (e.g., glycol, CO₂, etc.) can be used in place of or in addition to water to serve thermal energy loads. In other embodiments, subplants **202-212** may provide heating and/or cooling directly to the building or campus without requiring an intermediate heat transfer fluid. These and other variations to waterside system **200** are within the teachings of the present invention.

[0046] Each of subplants **202-212** can include a variety of equipment configured to facilitate the functions of the subplant. For example, heater subplant **202** is shown to include a plurality of heating elements **220** (e.g., boilers, electric heaters, etc.) configured to add heat to the hot water in hot water loop **214**. Heater subplant **202** is also shown to include several pumps **222** and **224** configured to circulate the hot water in hot water loop **214** and to control the flow rate of the hot water through individual heating elements **220**. Chiller subplant **206** is shown to include a plurality of chillers **232** configured to remove heat from the cold water in cold water loop **216**. Chiller subplant **206** is also shown to include several pumps **234** and **236** configured to circulate the cold water in cold water loop **216** and to control the flow rate of the cold water through individual chillers **232**.

[0047] Heat recovery chiller subplant **204** is shown to include a plurality of heat recovery heat exchangers **226** (e.g., refrigeration circuits) configured to transfer heat from cold water loop **216** to hot water loop **214**. Heat recovery chiller subplant **204** is also shown to include several pumps **228** and **230** configured to circulate the hot water and/or cold water through heat recovery heat exchangers **226** and to control the flow rate of the water through individual heat recovery heat exchangers **226**. Cooling tower subplant **208** is shown to include a plurality of cooling towers **238** configured to remove heat from the condenser water in condenser water loop **218**. Cooling tower subplant **208** is also shown to include several pumps **240** configured to circulate the condenser water in condenser water loop **218** and to control the flow rate of the condenser water through individual cooling towers **238**.

[0048] Hot TES subplant **210** is shown to include a hot TES tank **242** configured to store the hot water for later use. Hot TES subplant **210** may also include one or more pumps or valves configured to control the flow rate of the hot water into or out of hot TES tank **242**. Cold TES subplant **212** is shown to include cold TES tanks **244** configured to store the cold water for later use. Cold TES subplant **212** may also include one or more pumps or valves configured to control the flow rate of the cold water into or out of cold TES tanks **244**.

[0049] In some embodiments, one or more of the pumps in waterside system **200** (e.g., pumps **222**, **224**, **228**, **230**, **234**, **236**, and/or **240**) or pipelines in waterside system **200** include an isolation valve associated therewith. Isolation valves can be integrated with the pumps or positioned upstream or downstream of the pumps to control the fluid flows in waterside system **200**. In various embodiments, waterside system **200** can include more, fewer, or different types of devices and/or subplants based on the particular configuration of waterside system **200** and the types of loads served by waterside system **200**.

Airside System **300**

[0050] Referring now to FIG. 3, a block diagram of an airside system **300** is shown, according to some embodiments. In various embodiments, airside system **300** may supplement or replace airside system **130** in HVAC system **100** or can be implemented separate from HVAC system **100**. When implemented in HVAC system **100**, airside system **300** can include a subset of the HVAC devices in HVAC system **100** (e.g., AHU **106**, VAV units **116**, ducts **112-114**, fans, dampers, etc.) and can be located in or around building **10**. Airside system **300** may operate to heat or cool an airflow provided to building **10** using a heated or chilled fluid provided by waterside system **200**.

[0051] In FIG. 3, airside system **300** is shown to include an economizer-type air handling unit (AHU) **302**. Economizer-type AHUs vary the amount of outside air and return air used by the air handling unit for heating or cooling. For example, AHU **302** may receive return air **304** from building zone **306** via return air duct **308** and may deliver supply air **310** to building zone **306** via supply air duct **312**. In some embodiments, AHU **302** is a rooftop unit located on the roof of building **10** (e.g., AHU **106** as shown in FIG. 1) or otherwise positioned to receive both return air **304** and outside air **314**. AHU **302** can be configured to operate exhaust air damper **316**, mixing damper **318**, and outside air damper **320** to control an amount of outside air **314** and return air **304** that combine to form supply air **310**. Any return air **304** that does not pass through mixing damper **318** can be exhausted from AHU **302** through exhaust damper **316** as exhaust air **322**.

[0052] Each of dampers **316-320** can be operated by an actuator. For example, exhaust air damper **316** can be operated by actuator **324**, mixing damper **318** can be operated by actuator **326**, and outside air damper **320** can be operated by actuator **328**. Actuators **324-328** may communicate with an AHU controller **330** via a communications link **332**. Actuators **324-328** may receive control signals from AHU controller **330** and may provide feedback signals to AHU controller **330**. Feedback signals can include, for example, an indication of a current actuator or damper position, an amount of torque or force exerted by the actuator, diagnostic information (e.g., results of diagnostic tests performed by actuators **324-328**), status information, commissioning information, configuration settings, calibration data, and/or other types of information or data that can be collected, stored, or used by actuators **324-328**. AHU controller **330** can be an economizer controller configured to use one or more control algorithms (e.g., state-based algorithms, extremum seeking control (ESC) algorithms, proportional-integral (PI) control algorithms, proportional-integral-derivative (PID) control algorithms, model predictive control (MPC) algorithms, feedback control algorithms, etc.) to control actuators **324-328**.

[0053] Still referring to FIG. 3, AHU **302** is shown to include a cooling coil **334**, a heating coil **336**, and a fan **338** positioned within supply air duct **312**. Fan **338** can be configured to force supply air **310** through cooling coil **334** and/or heating coil **336** and provide supply air **310** to building zone **306**. AHU controller **330** may communicate with fan **338** via communications link **340** to control a flow rate of supply air **310**. In some embodiments, AHU controller **330** controls an amount of heating or cooling applied to supply air **310** by modulating a speed of fan **338**.

[0054] Cooling coil **334** may receive a chilled fluid from waterside system **200** (e.g., from cold water loop **216**) via piping **342** and may return the chilled fluid to waterside system **200** via piping **344**. Valve **346** can be positioned along piping **342** or piping **344** to control a flow rate of the chilled fluid through cooling coil **334**. In some embodiments, cooling coil **334** includes multiple stages of cooling coils that can be independently activated and deactivated (e.g., by AHU controller **330**, by BMS controller **366**, etc.) to modulate an amount of cooling applied to supply air **310**.

[0055] Heating coil **336** may receive a heated fluid from waterside system **200** (e.g., from hot water loop **214**) via piping **348** and may return the heated fluid to waterside system **200** via piping **350**. Valve **352** can be positioned along piping **348** or piping **350** to control a flow rate of the heated fluid through heating coil **336**. In some embodiments, heating coil **336** includes multiple stages of heating coils that can be independently activated and deactivated (e.g., by AHU controller **330**, by BMS controller **366**, etc.) to modulate an amount of heating applied to supply air **310**.

[0056] Each of valves **346** and **352** can be controlled by an actuator. For example, valve **346** can be controlled by actuator **354** and valve **352** can be controlled by actuator **356**. Actuators **354-356** may communicate with AHU controller **330** via communications links **358-360**. Actuators **354-356** may receive control signals from AHU controller **330** and may provide feedback signals to controller **330**. In some embodiments, AHU controller **330** receives a measurement of the supply air temperature from a temperature sensor **362** positioned in supply air duct **312** (e.g., downstream of cooling coil **334** and/or heating coil **336**). AHU controller **330** may also receive a measurement of the temperature of building zone **306** from a temperature sensor **364** located in building zone **306**.

[0057] In some embodiments, AHU controller **330** operates valves **346** and **352** via actuators **354-356** to modulate an amount of heating or cooling provided to supply air **310** (e.g., to achieve a setpoint temperature for supply air **310** or to maintain the temperature of supply air **310** within a setpoint temperature range). The positions of valves **346** and **352** affect the amount of heating or cooling provided to supply air **310** by cooling coil **334** or heating coil **336** and may correlate with the amount of energy consumed to achieve a desired supply air temperature. AHU **330** may control the temperature of supply air **310** and/or building zone **306** by activating or deactivating coils **334-336**, adjusting a speed of fan **338**, or a

combination of both.

[0058] Still referring to FIG. 3, airside system **300** is shown to include a building management system (BMS) controller **366** and a client device **368**. BMS controller **366** can include one or more computer systems (e.g., servers, supervisory controllers, subsystem controllers, etc.) that serve as system level controllers, application or data servers, head nodes, or master controllers for airside system **300**, waterside system **200**, HVAC system **100**, and/or other controllable systems that serve building **10**. BMS controller **366** may communicate with multiple downstream building systems or subsystems (e.g., HVAC system **100**, a security system, a lighting system, waterside system **200**, etc.) via a communications link **370** according to like or disparate protocols (e.g., LON, BACnet, etc.). In various embodiments, AHU controller **330** and BMS controller **366** can be separate (as shown in FIG. 3) or integrated. In an integrated implementation, AHU controller **330** can be a software module configured for execution by a processor of BMS controller **366**.

[0059] In some embodiments, AHU controller **330** receives information from BMS controller **366** (e.g., commands, setpoints, operating boundaries, etc.) and provides information to BMS controller **366** (e.g., temperature measurements, valve or actuator positions, operating statuses, diagnostics, etc.). For example, AHU controller **330** may provide BMS controller **366** with temperature measurements from temperature sensors **362-364**, equipment on/off states, equipment operating capacities, and/or any other information that can be used by BMS controller **366** to monitor or control a variable state or condition within building zone **306**.

[0060] Client device **368** can include one or more human-machine interfaces or client interfaces (e.g., graphical user interfaces, reporting interfaces, text-based computer interfaces, client-facing web services, web servers that provide pages to web clients, etc.) for controlling, viewing, or otherwise interacting with HVAC system **100**, its subsystems, and/or devices. Client device **368** can be a computer workstation, a client terminal, a remote or local interface, or any other type of user interface device. Client device **368** can be a stationary terminal or a mobile device. For example, client device **368** can be a desktop computer, a computer server with a user interface, a laptop computer, a tablet, a smartphone, a PDA, or any other type of mobile or non-mobile device. Client device **368** may communicate with BMS controller **366** and/or AHU controller **330** via communications link **372**.

Building Management System **400**

[0061] Referring now to FIG. 4, a block diagram of a building management system (BMS) **400** is shown, according to some embodiments. BMS **400** can be implemented in building **10** to automatically monitor and control various building functions. BMS **400** is shown to include BMS controller **366** and a plurality of building subsystems **428**. Building subsystems **428** are shown to include a building electrical subsystem **434**, an information communication technology (ICT) subsystem **436**, a security subsystem **438**, a HVAC subsystem **440**, a lighting subsystem **442**, a lift/escalators subsystem **432**, and a fire safety subsystem **430**. In various embodiments, building subsystems **428** can include fewer, additional, or alternative subsystems. For example, building subsystems **428** may also or alternatively include a refrigeration subsystem, an advertising or signage subsystem, a cooking subsystem, a vending subsystem, a printer or copy service subsystem, or any other type of building subsystem that uses controllable equipment and/or sensors to monitor or control building **10**. In some embodiments, building subsystems **428** include waterside system **200** and/or airside system **300**, as described with reference to FIGS. 2 and 3.

[0062] Each of building subsystems **428** can include any number of devices, controllers, and connections for completing its individual functions and control activities. HVAC subsystem **440** can include many of the same components as HVAC system **100**, as described with reference to FIGS. 1-3. For example, HVAC subsystem **440** can include a chiller, a boiler, any number of air handling units, economizers, field controllers, supervisory controllers, actuators, temperature sensors, thermostats, and other devices for controlling the temperature, humidity, airflow, or other variable conditions within building **10**. Lighting subsystem **442** can include any number of light fixtures, ballasts, lighting sensors, dimmers, and/or other devices configured to controllably adjust the amount of light provided to a building space. Security subsystem **438** can include occupancy sensors, video surveillance cameras, digital video recorders, video processing servers, intrusion detection devices, access control devices and servers, and/or other security-related devices.

[0063] Still referring to FIG. 4, BMS controller **366** is shown to include a communications interface **407** and a BMS interface **409**. Communications interface **407** may facilitate communications between BMS controller **366** and external applications (e.g., monitoring and reporting applications **422**, enterprise control applications **426**, remote systems and applications **444**, applications residing on client devices **448**, etc.) for allowing user control, monitoring, and adjustment to BMS controller **366** and/or subsystems **428**. Communications interface **407** may also facilitate communications between BMS controller **366** and client devices **448**. BMS interface **409** may facilitate communications between BMS controller **366** and building subsystems **428** (e.g., HVAC, lighting security, lifts, power distribution, business, etc.).

[0064] Communications interfaces **407** and/or BMS interface **409** can be or include wired or wireless communications interfaces (e.g., jacks, antennas, transmitters, receivers, transceivers, wire terminals, etc.) for conducting data communications with building subsystems **428** or other external systems or devices. In various embodiments, communications via communications interfaces **407** and/or BMS interface **409** can be direct (e.g., local wired or wireless communications) or via a communications network **446** (e.g., a WAN, the Internet, a cellular network, etc.). For example, communications interfaces **407** and/or BMS interface **409** can include an Ethernet card and port for sending and receiving data via an Ethernet-based communications link or network. In another example, communications interfaces **407** and/or BMS interface **409** can include a Wi-Fi transceiver for communicating via a wireless communications network. In another example, one or both of communications interfaces **407** and BMS interface **409** can include cellular or mobile phone communications transceivers. In one embodiment, communications interface **407** is a power line communications interface and BMS interface **409** is an Ethernet interface. In other embodiments, both communications interface **407** and BMS interface **409** are Ethernet interfaces or are the same Ethernet interface.

[0065] Still referring to FIG. 4, BMS controller **366** is shown to include a processing circuit **404** including a processor **406** and memory **408**. Processing circuit **404** can be communicably connected to BMS interface **409** and/or communications interface **407** such that processing circuit **404** and the various components thereof can send and receive data via communications interfaces **407** and/or BMS interface **409**. Processor **406** can be implemented as a general purpose processor, an application specific integrated circuit (ASIC), one or more field programmable gate arrays (FPGAs), a group of processing components, or other suitable electronic processing components.

[0066] Memory **408** (e.g., memory, memory unit, storage device, etc.) can include one or more devices (e.g., RAM, ROM, Flash memory, hard disk storage, etc.) for storing data and/or computer code for completing or facilitating the various processes, layers and modules described in the present application. Memory **408** can be or include volatile memory or non-volatile memory. Memory **408** can include database components, object code components, script components, or any other type of information structure for supporting the various activities and information structures described in the present application. According to some embodiments, memory **408** is communicably connected to processor **406** via processing circuit **404** and includes computer code for executing (e.g., by processing circuit **404** and/or processor **406**) one or more processes described herein.

[0067] In some embodiments, BMS controller **366** is implemented within a single computer (e.g., one server, one housing, etc.). In various other embodiments BMS controller **366** can be distributed across multiple servers or computers (e.g., that can exist in distributed locations). Further, while FIG. 4 shows applications **422** and **426** as existing outside of BMS controller **366**, in some embodiments, applications **422** and **426** can be hosted within BMS controller **366** (e.g., within memory **408**).

[0068] Still referring to FIG. 4, memory **408** is shown to include an enterprise integration layer **410**, an automated measurement and validation (AM&V) layer **412**, a demand response (DR) layer **414**, a fault detection and diagnostics (FDD) layer **416**, an integrated control layer **418**, and a building subsystem integration later **420**. Layers **410-420** can be configured to receive inputs from building subsystems **428** and other data sources, determine optimal control actions for building subsystems **428** based on the inputs, generate control signals based on the optimal control actions, and provide the generated control signals to building subsystems **428**. The following paragraphs describe some of the general functions performed by

each of layers **410-420** in **BMS 400**.

[0069] Enterprise integration layer **410** can be configured to serve clients or local applications with information and services to support a variety of enterprise-level applications. For example, enterprise control applications **426** can be configured to provide subsystem-spanning control to a graphical user interface (GUI) or to any number of enterprise-level business applications (e.g., accounting systems, user identification systems, etc.). Enterprise control applications **426** may also or alternatively be configured to provide configuration GUIs for configuring BMS controller **366**. In yet other embodiments, enterprise control applications **426** can work with layers **410-420** to optimize building performance (e.g., efficiency, energy use, comfort, or safety) based on inputs received at communications interface **407** and/or BMS interface **409**.

[0070] Building subsystem integration layer **420** can be configured to manage communications between BMS controller **366** and building subsystems **428**. For example, building subsystem integration layer **420** may receive sensor data and input signals from building subsystems **428** and provide output data and control signals to building subsystems **428**. Building subsystem integration layer **420** may also be configured to manage communications between building subsystems **428**. Building subsystem integration layer **420** translate communications (e.g., sensor data, input signals, output signals, etc.) across a plurality of multi-vendor/multi-protocol systems.

[0071] Demand response layer **414** can be configured to optimize resource usage (e.g., electricity use, natural gas use, water use, etc.) and/or the monetary cost of such resource usage in response to satisfy the demand of building **10**. The optimization can be based on time-of-use prices, curtailment signals, energy availability, or other data received from utility providers, distributed energy generation systems **424**, from energy storage **427** (e.g., hot TES **242**, cold TES **244**, etc.), or from other sources. Demand response layer **414** may receive inputs from other layers of BMS controller **366** (e.g., building subsystem integration layer **420**, integrated control layer **418**, etc.). The inputs received from other layers can include environmental or sensor inputs (e.g., internal to building **10**, external to building **10**, etc.) such as temperature, carbon dioxide levels, relative humidity levels, air quality sensor outputs, occupancy sensor outputs, room schedules, weather conditions, and the like. The inputs may also include inputs such as electrical use (e.g., expressed in kWh), thermal load measurements, pricing information, projected pricing, smoothed pricing, curtailment signals from utilities, and the like.

[0072] According to some embodiments, demand response layer **414** includes control logic for responding to the data and signals it receives. These responses can include communicating with the control algorithms in integrated control layer **418**, changing control strategies, changing setpoints, or activating/deactivating building equipment or subsystems in a controlled manner. Demand response layer **414** may also include control logic configured to determine when to utilize stored energy. For example, demand response layer **414** may determine to begin using energy from energy storage **427** just prior to the beginning of a peak use hour.

[0073] In some embodiments, demand response layer **414** includes a control module configured to actively initiate control actions (e.g., automatically changing setpoints, etc.) which minimize energy costs based on one or more inputs representative of or based on demand (e.g., price, a curtailment signal, a demand level, etc.). In some embodiments, demand response layer **414** uses equipment models to determine an optimal set of control actions. The equipment models can include, for example, thermodynamic models describing the inputs, outputs, and/or functions performed by various sets of building equipment. Equipment models may represent collections of building equipment (e.g., subplants, chiller arrays, etc.) or individual devices (e.g., individual chillers, heaters, pumps, etc.).

[0074] Demand response layer **414** may further include or draw upon one or more demand response policy definitions (e.g., databases, XML files, etc.). The policy definitions can be edited or adjusted by a user (e.g., via a graphical user interface, etc.) so that the control actions initiated in response to demand inputs can be tailored for the user's application, desired comfort level, particular building equipment, and/or based on other concerns. For example, the demand response policy definitions can specify which equipment can be turned on or off in response to particular demand inputs, how long a system or piece of equipment should be turned off, what setpoints can be changed, what the allowable set point adjustment range is, how long to hold a high demand setpoint before returning to a normally scheduled setpoint, how close to approach capacity limits, which equipment modes to utilize, the energy transfer rates (e.g., the maximum rate, an alarm rate, other rate boundary information, etc.) into and out of energy storage devices (e.g., thermal storage tanks, battery banks, etc.), and/or when to dispatch on-site generation of energy (e.g., via fuel cells, a motor generator set, etc.).

[0075] Integrated control layer **418** can be configured to use the data input or output of building subsystem integration layer **420** and/or demand response later **414** to make control decisions. Due to the subsystem integration provided by building subsystem integration layer **420**, integrated control layer **418** can integrate control activities of the subsystems **428** such that the subsystems **428** behave as a single integrated supersystem. In some embodiments, integrated control layer **418** includes control logic that uses inputs and outputs from a plurality of building subsystems to provide greater comfort and energy savings relative to the comfort and energy savings that separate subsystems could provide alone. For example, integrated control layer **418** can be configured to use an input from a first subsystem to make an energy-saving control decision for a second subsystem. Results of these decisions can be communicated back to building subsystem integration layer **420**.

[0076] Integrated control layer **418** is shown to be logically below demand response layer **414**. Integrated control layer **418** can be configured to enhance the effectiveness of demand response layer **414** by enabling building subsystems **428** and their respective control loops to be controlled in coordination with demand response layer **414**. This configuration may advantageously reduce disruptive demand response behavior relative to conventional systems. For example, integrated control layer **418** can be configured to assure that a demand response-driven upward adjustment to the setpoint for chilled water temperature (or another component that directly or indirectly affects temperature) does not result in an increase in fan energy (or other energy used to cool a space) that would result in greater total building energy use than was saved at the chiller.

[0077] Integrated control layer **418** can be configured to provide feedback to demand response layer **414** so that demand response layer **414** checks that constraints (e.g., temperature, lighting levels, etc.) are properly maintained even while demanded load shedding is in progress. The constraints may also include setpoint or sensed boundaries relating to safety, equipment operating limits and performance, comfort, fire codes, electrical codes, energy codes, and the like. Integrated control layer **418** is also logically below fault detection and diagnostics layer **416** and automated measurement and validation layer **412**. Integrated control layer **418** can be configured to provide calculated inputs (e.g., aggregations) to these higher levels based on outputs from more than one building subsystem.

[0078] Automated measurement and validation (AM&V) layer **412** can be configured to verify that control strategies commanded by integrated control layer **418** or demand response layer **414** are working properly (e.g., using data aggregated by AM&V layer **412**, integrated control layer **418**, building subsystem integration layer **420**, FDD layer **416**, or otherwise). The calculations made by AM&V layer **412** can be based on building system energy models and/or equipment models for individual BMS devices or subsystems. For example, AM&V layer **412** may compare a model-predicted output with an actual output from building subsystems **428** to determine an accuracy of the model.

[0079] Fault detection and diagnostics (FDD) layer **416** can be configured to provide on-going fault detection for building subsystems **428**, building subsystem devices (i.e., building equipment), and control algorithms used by demand response layer **414** and integrated control layer **418**. FDD layer **416** may receive data inputs from integrated control layer **418**, directly from one or more building subsystems or devices, and/or from another data source. FDD layer **416** may automatically diagnose and respond to detected faults. The responses to detected or diagnosed faults can include providing an alert message to a user, a maintenance scheduling system, or a control algorithm configured to attempt to repair the fault or to work-around the fault.

[0080] FDD layer **416** can be configured to output a specific identification of the faulty component or cause of the fault (e.g., loose damper linkage, etc.) using detailed subsystem inputs available at building subsystem integration layer **420**. In other exemplary embodiments, FDD layer **416** is configured to provide "fault" events to integrated control layer **418** which executes control strategies and policies in response to the received fault events. According to some embodiments, FDD layer **416** (or a policy executed by an integrated control engine or business rules engine) may shut-

down systems or direct control activities around faulty devices or systems to reduce energy waste, extend equipment life, or assure proper control response.

[0081] FDD layer **416** can be configured to store or access a variety of different system data stores (or data points for live data). FDD layer **416** may use some content of the data stores to identify faults at the equipment level (e.g., specific chiller, specific AHU, specific terminal unit, etc.) and other content to identify faults at component or subsystem levels. For example, building subsystems **428** may generate temporal (i.e., time-series) data indicating the performance of BMS **400** and the various components thereof. The data generated by building subsystems **428** can include measured or calculated values that exhibit statistical characteristics and provide information about how the corresponding system or process (e.g., a temperature control process, a flow control process, etc.) is performing in terms of error from its setpoint. These processes can be examined by FDD layer **416** to expose when the system begins to degrade in performance and alert a user to repair the fault before it becomes more severe.

Building Management System **500**

[0082] Referring now to FIG. 5, a block diagram of another building management system (BMS) **500** is shown, according to some embodiments. BMS **500** can be used to monitor and control the devices of HVAC system **100**, waterside system **200**, airside system **300**, building subsystems **428**, as well as other types of BMS devices (e.g., lighting equipment, security equipment, etc.) and/or HVAC equipment. In some embodiments, the building management system includes a time varying performance indication system.

[0083] BMS **500** provides a system architecture that facilitates automatic equipment discovery and equipment model distribution. Equipment discovery can occur on multiple levels of BMS **500** across multiple different communications busses (e.g., a system bus **554**, zone buses **556-560** and **564**, sensor/actuator bus **566**, etc.) and across multiple different communications protocols. In some embodiments, equipment discovery is accomplished using active node tables, which provide status information for devices connected to each communications bus. For example, each communications bus can be monitored for new devices by monitoring the corresponding active node table for new nodes. When a new device is detected, BMS **500** can begin interacting with the new device (e.g., sending control signals, using data from the device) without user interaction. [0084] Some devices in BMS **500** present themselves to the network using equipment models. An equipment model defines equipment object attributes, view definitions, schedules, trends, and the associated BACnet value objects (e.g., analog value, binary value, multistate value, etc.) that are used for integration with other systems. Some devices in BMS **500** store their own equipment models. Other devices in BMS **500** have equipment models stored externally (e.g., within other devices). For example, a zone coordinator **508** can store the equipment model for a bypass damper **528**. In some embodiments, zone coordinator **508** automatically creates the equipment model for bypass damper **528** or other devices on zone bus **558**. Other zone coordinators can also create equipment models for devices connected to their zone busses. The equipment model for a device can be created automatically based on the types of data points exposed by the device on the zone bus, device type, and/or other device attributes. Several examples of automatic equipment discovery and equipment model distribution are discussed in greater detail below.

[0085] Still referring to FIG. 5, BMS **500** is shown to include a time varying performance indication system **502**, a system manager **503**; several zone coordinators **506, 508, 510** and **518**; and several zone controllers **524, 530, 532, 536, 548**, and **550**. System manager **503** can monitor various data points in BMS **500** and report monitored variables to time varying performance indication system **502**. System manager **503** can communicate with client devices **504** (e.g., user devices, desktop computers, laptop computers, mobile devices, etc.) via a data communications link **574** (e.g., BACnet IP, Ethernet, wired or wireless communications, etc.). System manager **503** can provide a user interface to client devices **504** via data communications link **574**. The user interface may allow users to monitor and/or control BMS **500** via client devices **504**.

[0086] In some embodiments, system manager **503** is connected with zone coordinators **506-510** and **518** via a system bus **554**. System manager **503** can be configured to communicate with zone coordinators **506-510** and **518** via system bus **554** using a master-slave token passing (MSTP) protocol or any other communications protocol. System bus **554** can also connect system manager **503** with other devices such as a constant volume (CV) rooftop unit (RTU) **512**, an input/output module (IOM) **514**, a thermostat controller **516** (e.g., a TEC5000 series thermostat controller), and a network automation engine (NAE) or third-party controller **520**. RTU **512** can be configured to communicate directly with system manager **503** and can be connected directly to system bus **554**. Other RTUs can communicate with system manager **503** via an intermediate device. For example, a wired input **562** can connect a third-party RTU **542** to thermostat controller **516**, which connects to system bus **554**.

[0087] System manager **503** can provide a user interface for any device containing an equipment model. Devices such as zone coordinators **506-510** and **518** and thermostat controller **516** can provide their equipment models to system manager **503** via system bus **554**. In some embodiments, system manager **503** automatically creates equipment models for connected devices that do not contain an equipment model (e.g., IOM **514**, third party controller **520**, etc.). For example, system manager **503** can create an equipment model for any device that responds to a device tree request. The equipment models created by system manager **503** can be stored within system manager **503**. System manager **503** can then provide a user interface for devices that do not contain their own equipment models using the equipment models created by system manager **503**. In some embodiments, system manager **503** stores a view definition for each type of equipment connected via system bus **554** and uses the stored view definition to generate a user interface for the equipment.

[0088] Each zone coordinator **506-510** and **518** can be connected with one or more of zone controllers **524, 530-532, 536**, and **548-550** via zone buses **556, 558, 560**, and **564**. Zone coordinators **506-510** and **518** can communicate with zone controllers **524, 530-532, 536**, and **548-550** via zone busses **556-560** and **564** using a MSTP protocol or any other communications protocol. Zone busses **556-560** and **564** can also connect zone coordinators **506-510** and **518** with other types of devices such as variable air volume (VAV) RTUs **522** and **540**, changeover bypass (COBP) RTUs **526** and **552**, bypass dampers **528** and **546**, and PEAK controllers **534** and **544**.

[0089] Zone coordinators **506-510** and **518** can be configured to monitor and command various zoning systems. In some embodiments, each zone coordinator **506-510** and **518** monitors and commands a separate zoning system and is connected to the zoning system via a separate zone bus. For example, zone coordinator **506** can be connected to VAV RTU **522** and zone controller **524** via zone bus **556**. Zone coordinator **508** can be connected to COBP RTU **526**, bypass damper **528**, COBP zone controller **530**, and VAV zone controller **532** via zone bus **558**. Zone coordinator **510** can be connected to PEAK controller **534** and VAV zone controller **536** via zone bus **560**. Zone coordinator **518** can be connected to PEAK controller **544**, bypass damper **546**, COBP zone controller **548**, and VAV zone controller **550** via zone bus **564**.

[0090] A single model of zone coordinator **506-510** and **518** can be configured to handle multiple different types of zoning systems (e.g., a VAV zoning system, a COBP zoning system, etc.). Each zoning system can include a RTU, one or more zone controllers, and/or a bypass damper. For example, zone coordinators **506** and **510** are shown as Verasys VAV engines (VVEs) connected to VAV RTUs **522** and **540**, respectively. Zone coordinator **506** is connected directly to VAV RTU **522** via zone bus **556**, whereas zone coordinator **510** is connected to a third-party VAV RTU **540** via a wired input **568** provided to PEAK controller **534**. Zone coordinators **508** and **518** are shown as Verasys COBP engines (VCEs) connected to COBP RTUs **526** and **552**, respectively. Zone coordinator **508** is connected directly to COBP RTU **526** via zone bus **558**, whereas zone coordinator **518** is connected to a third-party COBP RTU **552** via a wired input **570** provided to PEAK controller **544**.

[0091] Zone controllers **524, 530-532, 536**, and **548-550** can communicate with individual BMS devices (e.g., sensors, actuators, etc.) via sensor/actuator (SA) busses. For example, VAV zone controller **536** is shown connected to networked sensors **538** via SA bus **566**. Zone controller **536** can communicate with networked sensors **538** using a MSTP protocol or any other communications protocol. Although only one SA bus **566** is shown in FIG. 5, it should be understood that each zone controller **524, 530-532, 536**, and **548-550** can be connected to a different SA bus. Each SA bus can connect a zone controller with various sensors (e.g., temperature sensors, humidity sensors, pressure sensors, light sensors, occupancy sensors, etc.), actuators (e.g., damper actuators, valve actuators, etc.) and/or other types of controllable equipment (e.g., chillers, heaters, fans, pumps, etc.).

[0092] Each zone controller **524, 530-532, 536**, and **548-550** can be configured to monitor and control a different building zone. Zone controllers

524, 530-532, 536, and 548-550 can use the inputs and outputs provided via their SA buses to monitor and control various building zones. For example, a zone controller 536 can use a temperature input received from networked sensors 538 via SA bus 566 (e.g., a measured temperature of a building zone) as feedback in a temperature control algorithm. Zone controllers 524, 530-532, 536, and 548-550 can use various types of control algorithms (e.g., state-based algorithms, extremum seeking control (ESC) algorithms, proportional-integral (PI) control algorithms, proportional-integral-derivative (PID) control algorithms, model predictive control (MPC) algorithms, feedback control algorithms, etc.) to control a variable state or condition (e.g., temperature, humidity, airflow, lighting, etc.) in or around building 10.

Time Varying Performance Indication System of Generating Performance Index for Connected Equipment

[0093] Referring now to FIG. 6, a block diagram of another building management system (BMS) 600 which includes the time varying performance indication system for generating a performance index for connected equipment is shown, according to some embodiments. BMS 600 can include many of the same components as BMS 400 and BMS 500 as described with reference to FIGS. 4 and 5. For example, BMS 600 is shown to include building 10, network 446, client devices 448, and time varying performance indication system 502. Building 10 is shown to include connected equipment 610, which can include any type of equipment used to monitor and/or control building 10. Connected equipment 610 can include connected chillers 612, connected AHUs 614, connected actuators 616, connected controllers 618, or any other type of equipment in a building HVAC system (e.g., boilers, economizers, valves, dampers, cooling towers, fans, pumps, etc.) or building management system (e.g., lighting equipment, security equipment, refrigeration equipment, etc.). Connected equipment 610 can include any of the equipment of HVAC system 100, waterside system 200, airside system 300, BMS 400, and/or BMS 500, as described with reference to FIGS. 1-5.

[0094] Connected equipment 610 can be outfitted with sensors to monitor particular conditions of the connected equipment 610. For example, chillers 612 can include sensors configured to monitor chiller variables such as chilled water return temperature, chilled water supply temperature, chilled water flow status (e.g., mass flow rate, volume flow rate, etc.), condensing water return temperature, condensing water supply temperature, motor amperage (e.g., of a compressor, etc.), variable speed drive (VSD) output frequency, and refrigerant properties (e.g., refrigerant pressure, refrigerant temperature, condenser pressure, evaporator pressure, etc.) at various locations in the refrigeration circuit. An example of a chiller 700 which can be used as one of chillers 612 is described in greater detail with reference to FIG. 7. Similarly, AHUs 614 can be outfitted with sensors to monitor AHU variables such as supply air temperature and humidity, outside air temperature and humidity, return air temperature and humidity, chilled fluid temperature, heated fluid temperature, damper position, etc. In general, connected equipment 610 monitor and report variables that characterize the performance of the connected equipment 610. Each monitored variable can be forwarded to network control engine 608 as a data point (e.g., including a point ID, a point value, etc.).

[0095] Monitored variables can include any measured or calculated values indicating the performance of connected equipment 610 and/or the components thereof. For example, monitored variables can include one or more measured or calculated temperatures (e.g., refrigerant temperatures, cold water supply temperatures, hot water supply temperatures, supply air temperatures, zone temperatures, etc.), pressures (e.g., evaporator pressure, condenser pressure, supply air pressure, etc.), flow rates (e.g., cold water flow rates, hot water flow rates, refrigerant flow rates, supply air flow rates, etc.), valve positions, resource consumptions (e.g., power consumption, water consumption, electricity consumption, etc.), control setpoints, model parameters (e.g., regression model coefficients, etc.), and/or any other time-series values that provide information about how the corresponding system, device, and/or process is performing. Monitored variables can be received from connected equipment 610 and/or from various components thereof. For example, monitored variables can be received from one or more controllers (e.g., BMS controllers, subsystem controllers, HVAC controllers, subplant controllers, AHU controllers, device controllers, etc.), BMS devices (e.g., chillers, cooling towers, pumps, heating elements, etc.), and/or collections of BMS devices.

[0096] Connected equipment 610 can also report equipment status information. Equipment status information can include, for example, the operational status of the equipment, an operating mode (e.g., low load, medium load, high load, etc.), an indication of whether the equipment is running under normal or abnormal conditions, a safety fault code, and/or any other information that indicates the current status of connected equipment 610. In some embodiments, equipment status information reported by the connected equipment 610 is in the form of status codes. For example, four types of status codes can be reported by a connected equipment (e.g., chiller), including safety shutdown codes (safety codes), warning codes, cycling codes, and operation codes. The status codes are described in greater detail herein below in this disclosure.

[0097] In some embodiments, each device of connected equipment 610 includes a control panel (e.g., control panel 710 shown in FIG. 7). The control panel can use the sensor data to shut down the device if the control panel determines that the device is operating under unsafe conditions. For example, the control panel can compare the sensor data (or a value derived from the sensor data) to predetermined thresholds. If the sensor data or calculated value crosses a safety threshold, the control panel can shut down the device and/or operate the device at a derated setpoint. The control panel can generate a data point when a safety shut down or a derate occurs. The data point can include a safety fault code which indicates the reason or condition that triggered the shut down or derate.

[0098] Connected equipment 610 can provide monitored variables and equipment status information to a network control engine 608. Network control engine 608 can include a building controller (e.g., BMS controller 366), a system manager (e.g., system manager 503), a network automation engine (e.g., NAE 520), or any other system or device of building 10 configured to communicate with connected equipment 610. In some embodiments, the monitored variables and the equipment status information are provided to network control engine 608 as data points. Each data point can include a point ID and/or a point value. The point ID can identify the type of data point and/or a variable measured by the data point (e.g., condenser pressure, refrigerant temperature, fault code, etc.). Monitored variables can be identified by name or by an alphanumeric code (e.g., Chilled_Water_Temp, 7694, etc.). The point value can include an alphanumeric value indicating the current value of the data point (e.g., 44° F., fault code 4, etc.).

[0099] Network control engine 608 can broadcast the monitored variables and the equipment status information to a remote operations center (ROC) 602. ROC 602 can provide remote monitoring services and can send an alert to building 10 in the event of a critical alarm. ROC 602 can push the monitored variables and equipment status information to a reporting database 604, where the data is stored for reporting and analysis. Time varying performance indication system 502 can access database 604 to retrieve the monitored variables and the equipment status information.

[0100] In some embodiments, time varying performance indication system 502 is a component of BMS controller 366 (e.g., within FDD layer 416). For example, time varying performance indication system 502 can be implemented as part of a METASYS® brand building automation system, as sold by Johnson Controls Inc. In other embodiments, time varying performance indication system 502 can be a component of a remote computing system or cloud-based computing system configured to receive and process data from one or more building management systems. For example, time varying performance indication system 502 can connect the connected equipment 610 (e.g., chillers 612) to the cloud and collect real-time data for over a number of points (e.g., 50 points) on those equipment. In other embodiments, time varying performance indication system 502 can be a component of a subsystem level controller (e.g., a HVAC controller, etc.), a subplant controller, a device controller (e.g., AHU controller 330, a chiller controller, etc.), a field controller, a computer workstation, a client device, and/or any other system and/or device that receives and processes monitored variables from connected equipment 610.

[0101] Time varying performance indication system 502 may use the monitored variables to identify a current operating state of connected equipment 610. The current operating state can be examined by time varying performance indication system 502 to expose when connected equipment 610 begins to degrade in performance and/or to predict when faults will occur. In some embodiments, time varying performance indication system 502 determines whether the current operating state is a normal operating state or a faulty operating state. Time varying performance indication system 502 may report the current operating state and/or the predicted faults to client devices 448, service technicians 606, building 10, and/or any other system and/or device. Communications between time varying performance indication system 502 and other systems

and/or devices can be direct and/or via an intermediate communications network, such as network **446**. If the current operating state is identified as a faulty state or moving toward a faulty state, time varying performance indication system **502** may generate an alert or notification for service technicians **606** to repair the fault or potential fault before it becomes more severe. In some embodiments, time varying performance indication system **502** uses the current operating state to determine an appropriate control action for connected equipment **610**.

[0102] In some embodiments, time varying performance indication system **502** provides a web interface which can be accessed by service technicians **606**, client devices **448**, and other systems or devices. The web interface can be used to access the raw data in reporting database **604**, view the results produced by the time varying performance indication system, identify which equipment is in need of preventative maintenance, and otherwise interact with time varying performance indication system **502**. Service technicians **606** can access the web interface to view a list of equipment for which faults are predicted by time varying performance indication system **502**. Service technicians **606** can use the predicted faults to proactively repair connected equipment **610** before a fault and/or an unexpected shut down occurs. These and other features of time varying performance indication system **502** are described in greater detail below.

[0103] Referring now to FIG. 7, a schematic diagram of a chiller **700** is shown, according to some embodiments. Chiller **700** is an example of a type of connected equipment **610** which can report monitored variables and status information (status codes) to time varying performance indication system **502**. Chiller **700** is shown to include a refrigeration circuit having a condenser **702**, an expansion valve **704**, an evaporator **706**, a compressor **708**, and a control panel **710**. In some embodiments, chiller **700** includes sensors that measure a set of monitored variables at various locations along the refrigeration circuit. Table 1 describes an exemplary set of monitored parameters/variables that can be measured in chiller **700**. Time varying performance indication system **502** can use these or other variables to detect the current operating state of chiller **700**, detect faults, predict potential/future faults, and/or determine diagnoses. Time varying performance indication system **502** may additionally use external parameters such as weather conditions and geographical location where the chiller **700** is operating.

TABLE-US-00001

TABLE 1	Monitored Chiller Parameters	Number	ID	Description	
1	MOT-FLA	Motor full load amps	2	CHWR-T	Chilled water return temperature
3	CHWS-T	Chilled water supply temperature	4	COND-P	Condenser pressure
5	EVAP-P	Evaporator pressure	6	CWR-T	Condensed water return temperature
7	CWS-T	Condensed water supply temperature	8	MTAMP-SP	Motor amps setpoint
9	CHWT-SP	Chilled water supply temperature setpoint	10	VFD OP-Hz	Variable frequency drive output frequency
11	CHWF-ST	Chilled water flow status			

[0104] Chiller **700** can be configured to operate in multiple different operating states. For example, chiller **700** can be operated in a low load state, a medium load state, a high load state, and/or various states therebetween. The operating states may represent the normal operating states or conditions of chiller **700**. Faults in chiller **700** may cause the operation of chiller **700** to deviate from the normal operating states. For example, various types of faults may occur in each of the normal operating states. For example, faults can be caused by stalling or surging in the compressor or other mechanical effects that can occur during operation. In some embodiments, time varying performance indication system **502** can collect or receive samples of the monitored variables. For example, system **502** may collect or receive 1000 samples of the monitored variables at a rate of one sample per second.

[0105] Referring now to FIG. 8, a block diagram illustrating the time varying performance indication system **502** in greater detail is shown, according to some embodiments. Time varying performance indication system **502** is shown to include a communications interface **810** and a processing circuit **812**. Communications interface **810** may facilitate communications between time varying performance indication system **502** and various external systems or devices. For example, time varying performance indication system **502** may receive the monitored variables from connected equipment **610** and provide control signals, performance indices, and/or other information of detected faults to connected equipment **610** via communications interface **810**. Communications interface **810** may also be used to communicate with remote systems and applications **444**, client devices **448**, and/or any other external system or device. For example, time varying performance indication system **502** may provide performance indices and other information of detected faults to remote systems and applications **444**, client devices **448**, service technicians **606**, or any other external system or device via communications interface **810**.

[0106] Communications interface **810** can include any number and/or type of wired or wireless communications interfaces (e.g., jacks, antennas, transmitters, receivers, transceivers, wire terminals, etc.). For example, communications interface **810** can include an Ethernet card and port for sending and receiving data via an Ethernet-based communications link or network. As another example, communications interface **810** can include a WiFi transceiver, a NFC transceiver, a cellular transceiver, a mobile phone transceiver, or the like for communicating via a wireless communications network. In some embodiments, communications interface **810** includes RS232 and/or RS485 circuitry for communicating with BMS devices (e.g., chillers, controllers, etc.). Communications interface **810** can be configured to use any of a variety of communications protocols (e.g., BACNet, Modbus, N2, MSTP, Zigbee, etc.). Communications via interface **810** can be direct (e.g., local wired or wireless communications) or via an intermediate communications network **446** (e.g., a WAN, the Internet, a cellular network, etc.). Communications interface **810** can be communicably connected with processing circuit **812**, and the various components thereof can send and receive data via communications interface **810**.

[0107] Processing circuit **812** is shown to include a processor **814** and memory **816**. Processor **814** can be implemented as a general purpose processor, an application specific integrated circuit (ASIC), one or more field programmable gate arrays (FPGAs), a group of processing components, or other suitable electronic processing components. Memory **816** (e.g., memory, memory unit, storage device, etc.) can include one or more devices (e.g., RAM, ROM, Flash memory, hard disk storage, etc.) for storing data and/or computer code for completing or facilitating the various processes, layers and modules described in the present application. Memory **816** can be or include volatile memory or non-volatile memory. Memory **816** can include database components, object code components, script components, or any other type of information structure for supporting the various activities and information structures described in the present application. According to some embodiments, memory **816** is communicably connected to processor **814** via processing circuit **812** and includes computer code for executing (e.g., by processing circuit **812** and/or processor **814**) one or more processes described herein.

[0108] Still referring to FIG. 8, in some embodiments, the memory **816** can include at least an input processing module **820**, a performance check module **822**, an individual performance check indicator generation module **824**, an overall performance index generation module **826**, a first weights determination module **828**, and a second weights determination module **830**. In other embodiments, more, less, or different modules or components can be stored in memory **816**. In some embodiments, the modules **820-830** can be implemented in one apparatus. In other embodiments, each of the modules **820-830** can be implemented in different and separate apparatuses and/or executed by different and separate processors, or a combination thereof. In some embodiments, modules **820-830** stored in a non-transitory computer readable medium (e.g., memory **816**) can be executed by the processor **814** to perform operations as described herein. In some embodiments, each of the modules **820-830** or a combination of some of the modules **820-830** can be implemented as hardware circuits.

[0109] Referring now to FIG. 9, a high level flow diagram illustrating a process **900** of generating a performance index for connected equipment is shown, according to some embodiments. In some embodiments, at stage **902**, the time varying performance indication system **502** can be configured to obtain time series data from past N time units and connected equipment specific design parameters. For example, the input processing module **820** can be configured to obtain time series data and connected equipment specific design parameters. The time units can be days, hours, minutes, seconds, weeks, months, or years. N is a number, such as an integer. For example, the past N time units can be the past 5 days, past 2 weeks, etc. In some embodiments, the time series data can include data points of a plurality of monitored variables and a plurality of status codes from the connected equipment **610**.

[0110] In some embodiments, the connected equipment **610** can be configured to measure a plurality of monitored variables and generate a plurality of status codes. As discussed herein above in relation to FIGS. 6 and 7, connected equipment **610** (e.g., chillers **612**, **700**) can measure monitored variables (e.g., measured or calculated temperatures, pressures, flow rates, valve positions, resource consumptions, control setpoints, model

parameters that can be any time-series values providing information about how the corresponding system, device, and/or process is performing. Connected equipment **610** can also provide or generate equipment status information in the form of status codes. In some embodiments, four types of status codes can be provided and reported by connected equipment (e.g., chiller), including safety shutdown codes (safety codes), warning codes, cycling codes, and operation codes. In the descriptions herein below, a chiller (e.g., chiller **612**, **700**) is used as an example of the connected equipment **610**. It should be understood that connected equipment is not limited to chillers and the operations described herein below can be performed for any connected equipment.

[0111] In some embodiments, safety shutdown codes are generated when safety shutdowns occur. Safety shutdowns can be triggered when certain conditions that are deemed dangerous to a chiller occur. These conditions may cause physical damage to the evaporator, condenser, compressor, variable speed drive (VSD), motor, or other components of the chiller. By the time a safety shutdown occurs, the chiller may have already sustained some damage. In some embodiments, depending on what causes the safety shutdown, it may require time and money to do a shutdown and machine servicing, or it could just require a reset of a chiller panel or building control strategy. In some embodiments, knowing the type of safety shutdown may not be sufficient to determine the root cause and solution. In some embodiments, warning codes do not shut down the chiller but give alerts that the chiller is not operating under a good condition. In some embodiments, cycling codes generally shut down a chiller due to specific conditions that occur in the chiller. For example, if a pump that feeds the condenser fails, the chiller may shut down due to loss of condenser flow. In some embodiments, operation codes indicate if the chiller is running, not running, or in alarm or shutdown states. In some embodiments, there are a number of different safety codes, warning codes, and cycling codes that can occur. In some embodiments, the operation codes are limited to a maximum number (e.g., 15) and a subset (e.g., 3) represents states when the chiller is running.

[0112] In some embodiments, the input processing module **820** of the time varying performance indication system **502** can receive or obtain time series data (e.g., data points of the plurality of monitored variables and the plurality of status codes) from the connect equipment **610** through the communication interface **810** via the network **446**. In some embodiments, the communication interface **810** can obtain the time series data from the reporting database **604**. Table 2 shows an example of the time series data that can be used as inputs to the operations performed by the time varying performance indication system **502**. In some embodiments, the data is collected from sensors and is related to physical quantities in the chiller. For example, for legacy chillers, points may be sampled every 1, 5, or 15 minutes, in some embodiments. For other chillers (e.g., SCC chillers), points may be change-of-value, in some embodiments.

TABLE-US-00002 TABLE 2 Sample Input Data varname timestamp ACC OP HRS ACC SYS STRT CHWF-STS CHWP-STS CHWR-T 2018-07-09 11377.0 885.0 0.0 0.0 13.7 00:45:00 2018-07-09 11377.0 885.0 0.0 0.0 13.8 01:00:00 2018-07-09 11377.0 885.0 0.0 0.0 13.8 01:15:00 2018-07-09 11377.0 885.0 0.0 0.0 13.7 01:30:00 2018-07-09 11377.0 885.0 0.0 0.0 13.7 01:45:00 varname timestamp CHWS-T CHWT-SP COND-AP COND-P CSAT-T 2018-07-09 12.9 6.7 -10.499999 368.9 13.8 00:45:00 2018-07-09 12.9 6.7 -10.499999 369.6 13.8 01:00:00 2018-07-09 13.0 7.3 -10.40001 369.6 13.8 01:15:00 2018-07-09 12.8 7.4 -10.40001 370.3 13.8 01:30:00 2018-07-09 12.8 7.4 -10.30001 370.3 13.9 01:45:00 varname timestamp VSD OP-Hz VSD OP-V VSD PH A-C VSD PH B-C VSD PH C-C 2018-07-09 0.0 0.0 0.0 0.0 0.0 00:45:00 2018- Jul. 9 0.0 0.0 0.0 0.0 0.0 01:00:00 2018-07-09 0.0 0.0 0.0 0.0 0.0 01:15:00 2018-07-09 0.0 0.0 0.0 0.0 0.0 01:30:00 2018-07-09 0.0 0.0 0.0 0.0 0.0 01:45:00 varname timestamp VSD-CONVHS-T VSD-SURG-CONT VSDDC-V VSDIA-T WAR-CODE 2018-07-09 23.0 943.0 1.0 31.0 0.0 00:45:00 2018-07-09 23.0 943.0 1.0 31.0 0.0 01:00:00 2018-07-09 23.0 943.0 1.0 31.0 0.0 01:15:00 2018-07-09 23.0 943.0 1.0 31.0 0.0 01:30:00 2018-07-09 23.0 943.0 1.0 31.0 0.0 01:45:00

[0113] As illustrated in Table 2 above, each input data or time series data can include a value and a timestamp indicating the time that the data is collected. For example, chilled water flow status (varname or ID: CHWP-STS) for this particular chiller has a value of 0.0 at the time 2018-07-09 00:45:00. It should be understood that the example input data as shown in Table 2 are for illustrative purposes only and should not be regarded as limiting in any way.

[0114] In some embodiments, the input processing module **820** of the time varying performance indication system **502** can obtain or receive connected equipment specific parameters. The connected equipment specific parameters are parameters specific to the connected equipment (e.g. chiller **612**, **700**). In some embodiments, the connected equipment specific parameters are obtained from the reporting database **604** via the communication interface **810**. In some embodiments, the connected equipment specific parameters are obtained from another system or storage via the network **446** through the communication interface **810**. In some embodiments, the connected equipment specific parameters are stored in a memory or local storage of the time varying performance indication system **502**. Example connected equipment specific parameters are illustrated with reference to Table 3 below.

[0115] Referring again to FIG. 9, in some embodiments, at stage **904**, the time varying performance indication system **502** can be configured to perform a plurality of performance checks for the connected equipment using the time series data (e.g., data points of the plurality of monitored variables and the plurality of status codes) and connected equipment specific parameters obtained or received at stage **902**. For example, the performance check module **822** can perform the performance checks for the connected equipment. In some embodiments, the performance checks include status checks (first performance checks) and health checks (second performance checks). In other embodiments, additional performance checks (e.g., raw sensor value checks, monitoring of long-term trends, setpoint deviations, vibration data, flow measurements, or any other checks with relevance to connected equipment health) can be performed for the connected equipment in addition to the status checks and health checks.

[0116] In some embodiments, the time varying performance indication system **502** (e.g., the performance check module **822**) can perform a plurality of status checks for the connected equipment using a plurality of status codes from the past N time units (e.g., past 5 days). In some embodiments, the performance check module **822** can identify safety shutdown codes (safety codes), warning codes, and cycling codes generated by the connected equipment. For example, the safety codes, warning codes, and cycling codes in the past 5 days can be identified by checking the timestamps associated with each code.

[0117] In some embodiments, the time varying performance indication system **502** (e.g., the performance check module **822**) can perform a plurality of health checks for the connected equipment using data points of the plurality of monitored variables from the past N time units (e.g., past 12 hours, past 5 days, past 2 weeks), a plurality of connected equipment specific parameters, and a plurality of predetermined rules. For example, the data points of the plurality of monitored variables from the past N time units and the plurality of connected equipment specific parameters obtained in stage **902** can be applied to a plurality of predetermined rules that are described in more detail below.

[0118] In some embodiments, the performance check module **822** can check a set of predetermined rules to determine if there is a violation of any of the rules. Responsive to a violation of one or more rules, the performance check module **822** can generate alerts or alarms depending on the degree of the severity of the violation and/or the rule being violated. Different connected equipment may have different health checks. In general, the health checks available for a certain connected equipment depend on the type of the connected equipment and customer configurations. Continuing using the example of the chiller as the connected equipment, in some embodiments, the performance check module **822** may consider a subset of the health checks that are used for the particular type of the chiller because not all health checks apply to every chiller. Table 3 shows a list of health check parameters (connected equipment specific parameters) and constants (thresholds) for a particular type of chiller as an illustrative example.

TABLE-US-00003 TABLE 3 Health check parameter and constant list for a type of chillers. Parameter Value Alert Alarm % FLA-MIN 10 Condenser Approach 3.5 5 VSD - High Internal Amb. Temp. 135 140 Low Refrigerant Level 15 10 Cond Ent. Water 85 Evaporator Approach 3.5 5 High Oil Temp. 155 165 Runtime Threshold - (HealthChart) 24 High Refrigerant Level 90 Low condenser water entering temp. 49 47 Runtime Threshold % - Condenser Approach 0.3 0.3 Runtime Threshold % - VSD - High 0.3 0.3 Internal Amb. temp Runtime Threshold % - Low Refrigerant Level 0.5 0.5 Runtime Threshold % - Cond Ent. Water 0.05 Runtime Threshold % - Evaporator Approach 0.3 0.3 Runtime Threshold % - High Oil Temp. 0.05 0.05 Runtime Threshold % - High Refrigerant Level 0.05 Runtime Threshold % - VSD Capacity Control 0.95 Alarm Runtime Threshold % - Low

condenser water 0.05 entering temp.

[0119] In some embodiments, for example, using the parameters and constants (thresholds) in Table 3, the performance check module **822** applies the following predetermined rules to perform the health checks. [0120] 1. High evaporator approach temperature: [0121] (MOT-FLA>Const (% FLA-MIN)) AND (EVAP-AP>Const (Evaporator Approach)) [0122] 2. High condenser approach temperature: [0123] (MOT-FLA>Const (% FLA-MIN)) AND (COND-AP>Const (Condenser Approach)) [0124] 3. High entering condenser water temperature: [0125] (MOT-FLA>Const (% FLA-MIN)) AND (CWS-T>Const (Cond Ent. Water)) [0126] 4. High condenser refrigerant level: [0127] (MOT-FLA>Const (% FLA-MIN)) AND (REF-POS>Const (High Refrigerant Level)) [0128] 5. Low condenser refrigerant level: [0129] (MOT-FLA>Const (% FLA-MIN)) AND (REF-POS<Const (Low Refrigerant Level)) AND (CHWR-T-CHWS-T>1.5) [0130] 6. High oil temperature while running: [0131] (MOT-FLA>Const (% FLA-MIN)) AND (OILS-T>Const (High Oil Temp.)) [0132] 7. Low entering condenser water temperature: [0133] (MOT-FLA>Const (% FLA-MIN)) AND (CWS-T<Const (Low condenser water entering temp.))

[0134] In the above example health check rules, evaluated data points are limited to those collected while the chiller was running. For example, these data points must have operation codes that correspond to “running” states (e.g., 8, 9, 12) and have the motor percent full load amps values above Const (% FLA-MIN). As shown in Table 3, the value of (% FLA-MIN)=10 in this example. In some embodiments, the number of these data points sets the “Run Time.” In some embodiments, “Alert Time” and “Alarm Time” indicate the number of data points where the chiller is running and is in alert and alarm conditions, respectively. For example, if the “Alert Time”/“Run Time” value exceeds the threshold listed in Table 3, the chiller is in Alert, and this health check is triggered. As an example, with respect to the first example health check rule for high evaporator approach temperature, an alert is triggered when the motor percent full load amps value (MOT-FLA) is above 10 and the evaporator approach (EVAP-AP) value is greater than 3.5. Continuing with this example, an alarm is triggered when the motor percent full load amps value (MOT-FLA) is above 10 and the evaporator approach (EVAP-AP) value is greater than 5.

[0135] In some embodiments, the time varying performance indication system **502** can determine a plurality of individual performance check indicators based on the status checks and the health checks using a plurality of first weights. For example, the individual performance check indicator generation module **824** can be configured to determine individual performance check indicators based on the status checks and the health checks. In some embodiments, the first weights are time based weights each determined based on a different timing. Example individual performance check indicators generation or determination processes are described in more detail in relation to FIGS. **11** and **12**.

[0136] Referring now to FIG. **11**, a flow diagram illustrating a process **1100** of generating a performance index for connected equipment is shown, according to some embodiments. In brief overview, a window of time series data can be provided as input, and a number of performance checks can be applied to it. In some embodiments, each of the performance checks returns a 0 if it passes, or a 1 if it fails. The values are then multiplied by their respective weights to obtain a penalty factor for each check. Performance check weights reflect the severity or impact of the problems they detect, and may also be time dependent if, for example, the recent events are weighted more heavily than events that took place several days ago. The penalty factors are then summed and subtracted from the maximum value of the index, I.sub.max (e.g., 10, 100) to obtain the value of the performance index. In some embodiments, while the index decreases with each performance check violation, it is constrained to stay above 0.

[0137] Referring to FIG. **11**, in further detail, at stage **1104**, a plurality (n numbers) of performance checks (e.g., status checks, health checks) are performed as described with reference to stage **904** of FIG. **9**. In some embodiments, an individual performance check indicator (e.g., f.sub.1,t, . . . f.sub.n,t) may be generated as a result of the respective performance check. For example, f.sub.n,t=0 if the respective performance check passes, and f.sub.n,t=1 if the respective performance check fails (e.g., an alert or alarm is triggered with respect to the health checks, a shutdown, warning, etc. are identified with respect to the status checks). In some embodiments, the individual performance check indicator is used with weights w.sub.1 . . . w.sub.n (second weights as described herein below) to generate the overall performance index. In the embodiments of the individual performance check indicator determination or generation process as shown in FIG. **11**, the individual performance check indicator (e.g., f.sub.1,t, . . . f.sub.n,t) has binary values (e.g., pass, fail). In other embodiments, the individual performance check indicator may have a series of values associated with time varying weights (first weights), as illustrated in relation to FIG. **12**.

[0138] Referring now to FIG. **12**, a flow diagram illustrating a process **1200** of generating a performance index for connected equipment is shown, according to some embodiments. FIG. **12** is similar to FIG. **11**, with a different embodiment for the individual performance check indicator (e.g., f.sub.1,t, . . . f.sub.n,t) generation or determination process. Referring to FIG. **12**, at stage **1204**, a plurality (n numbers) of performance checks (e.g., status checks, health checks) are performed as described with reference to stage **904** of FIG. **9**. In the embodiments of FIG. **12**, the individual performance check indicators (e.g., f.sub.1,t, . . . f.sub.n,t) are time dependent when data from a period of time or range of days are considered. As shown in stage **1204** of FIG. **12**, individual performance check indicators have different values based on when the events (e.g., an alert or alarm with respect to the health checks, a shutdown, warning, etc. are identified with respect to the status checks) occurred. For example, most recent events may be weighted heavier than earlier events during the past N time units (e.g., past 5 days). As shown in the examples of FIG. **12**, the individual performance check indicator (e.g., f.sub.1,t, . . . f.sub.n,t) decreases in value depending upon when the event last happened. In the embodiments of FIG. **12**, a set of time varying weights (first weights) 0, 0.14, 0.22, 0.367, 0.606, 1.0 are used for the individual performance check indicators. For example, f.sub.n,t=0 if no occurrence of the event in past 120 hours, f.sub.n,t=0.14 if last occurrence is between 96 and 120 hours, f.sub.n,t=0.22 if last occurrence is between 72 and 96 hours, f.sub.n,t=0.367 if last occurrence is between 48 and 72 hours, f.sub.n,t=0.606 if last occurrence is between 24 and 48 hours, and f.sub.n,t=1 if last occurrence is between 0 and 24 hours.

[0139] In some embodiments, the first weights determination module **828** can determine the first weights used for the individual performance check indicators. For example, the first weights can be the time varying weights used for the individual performance check indicators described above in relation to FIG. **12**. In some embodiments, the time varying weights are determined using the exponential decay function with a tau value of two. FIG. **13** shows a graph **1300** of the exponential decay function with a tau value of two, according to some embodiments. As shown in FIG. **13**, the X-axis represents time and the dots show values used. In other embodiments, different methods can be used to determine the first weights or the time varying weights.

[0140] Referring back to FIG. **9**, in some embodiments, at stage **906**, the time varying performance indication system **502** can be configured to generate an overall performance index for the connected equipment using the plurality of individual performance check indicators and a plurality of second weights. For example, the overall performance index generation module **826** can generate an overall performance index for the connected equipment. In some embodiments, an overall performance index in the form of a number index value (0-I.sub.max) is produced at stage **908**, where I.sub.max is a positive number (e.g., 10, 100), in some embodiments. FIGS. **10**, **11** and **12** include a more detailed illustration of the overall performance index generation.

[0141] Referring now to FIG. **10**, a flow diagram illustrating a process **1000** of generating a performance index for connected equipment is shown, according to some embodiments. Stages **1002**, **1004** and **1008** can be identical or similar to stages **902**, **904** and **908** as described with respect to FIG. **9** and will not be described here again. Referring to FIG. **10**, in some embodiments, at stage **1006**, a plurality of individual performance check indicators **1006a** and a plurality of second weights **1006b** are used to produce overall performance index. For example, the individual performance check indicators described herein above in relation to FIGS. **11** and **12** can be used to generate the overall performance index. In some embodiments, the overall performance index is calculated as the summation of the products of all of the performance checks (individual performance check indicators) and their corresponding severity weights, and then subtracting this value from the I.sub.max. In some embodiments, the I.sub.max is a positive number (e.g., 10, 100).

[0142] In some embodiments, the following equation (also as shown in stages **1106** and **1206** of FIGS. **11** and **12**) can be used to generate the overall performance index for the connected equipment.

$$I_t = \max(0, I_{\max} - \sum_{i=0}^n W_i f_{i,t}) \quad (1)$$

where $I_{\text{sub},t}$ is the value of the performance index at time t , $f_{\text{sub},i,t}$ represents the i th individual performance check indicator (performance check) of a total n number individual performance check indicators (performance checks) at time t , and $W_{\text{sub},i}$ represents the weight (second weight) for the corresponding (i th) individual performance check indicator fit. It should be understood that the performance index formula as shown above is only one of several possible implementations and should not be regarded as limiting in any way.

[0143] In some embodiments, the second weights determination module **830** can determine the second weights. In some embodiments, the second weights can be determined based on severity or impact of the type of events associated with the performance check. In some embodiments, the plurality of second weights can be severity weights each representing a predetermined degree of severity of a respective first performance check or a respective second performance check. For example, a safety shut down event (represented by a safety code) is more severe than a warning event (represented by a warning code), and a health check alarm is more severe than a health check alert. In some embodiments, the second weights can be exponentially or linearly decaying weights based on exponential decay function or linear decay function. Example severity weights are shown in table **1010** in FIG. **10**. As shown in table **1010**, each severity weight is a product of a weighting factor (e.g., 0.3, 0.1, 0.1, 0.1, 0.2) and $I_{\text{sub},\max}$. For example, when $I_{\text{sub},\max}=10$, the weight of the safety codes is 3, the weight of the warning code codes is 1, the weight of the cycling code is 1, the weight of the health check alert is 1, and the weight of the health check alarm is 2, in some embodiments. While the embodiment of table **1010** uses five severity weights that cover codes or checks for safety codes, cycling code, warning codes, health check alerts, and health check alarms, in other embodiments, each of these codes can be sort through and various weights can be assigned for individual codes and checks, increasing the number of severity weights overall.

[0144] The following is an example of calculating an overall performance index based on the embodiments of performance checks illustrated in FIG. **11** where time weighting (time varying weights) is not considered. This example illustrates how the overall performance index can be calculated for a daily run. For example, if for the first 5-day window, the status checks have 1 safety code, 2 warning codes, and 0 cycling codes detected, and the health checks have 1 health check alert and 0 alarms detected, then the overall performance index can be calculated as:

$$\text{PerformanceIndexforwindow}x = \max(0, 10 - [(\text{\#ofsafetycodes} * 3) + (\text{\#ofwarningcodes} * 1) + (\text{\#ofcyclingcodes} * 1) + (\text{\#ofhealthcheckalerts} * 1) + (\text{\#ofhealthcheckalarms} * 2)])$$

[0145] In the above example, the maximum value $I_{\text{sub},\max}=10$, and the overall performance index is constrained between 0 and $I_{\text{sub},\max}$, that is, between 0 and 10 in this example. In some embodiments, the above calculation can be repeated for each window through the data.

[0146] The following is an example of calculating an overall performance index based on the embodiments of performance checks illustrated in FIG. **12** where time weighting (time varying weights) is considered. This example is described in relation to FIG. **14**. Referring now to FIG. **14**, an example scenario **1400** of performance checks for generating an overall performance index is shown, according to some embodiments. As shown in FIG. **14**, performance checks are performed for the past 5 days. For example, health checks on “High Evap App” have detected 3 health check alerts (**1402**, **1404**, **1406**) and 2 health check alarms (**1408**, **1410**) in the past 5 days. Health checks on “Low Cond. Ref Level” have detected 1 health check alert (**1420**) occurred 72-96 hours ago. Similarly, status checks have detected 1 safety code (**1430**) on “Evap Low Pressure” occurred 96-120 hours ago, 1 cycling code (**1450**) on “Power Fault” 72-96 hours ago, 1 warning code (**1440**) on “High Oil Temp” 24-48 hours ago, and 2 warning codes (**1460**, **1462**) on “Low Oil Pressure” 96-120 hours ago and 48-72 hours ago.

[0147] In some embodiments, only the most recent instance of a performance check is considered in the calculation of the performance index. For example, as shown in FIG. **14**, for the health checks on “High Evap App,” only the most recent instance (**1410**) is considered in the calculation of the performance index. Similarly, since there are 2 warning codes (**1460**, **1462**) on “Low Oil Pressure,” only the most recent instance (**1462**) is considered. As shown in FIG. **14**, performance checks (**1410**, **1420**, **1430**, **1440**, **1450**, and **1462**) that are considered in the calculation of the performance index are circled.

[0148] Continuing with the example in FIG. **14**, in some embodiments, using equation (1) as described above, the overall performance index can be calculated as:

$$\text{PerformanceIndex} = \max(0, 10 - [(\text{sumofsafetycodes} * W_{\text{sc}}) + (\text{sumofwarningcodes} * W_{\text{wc}}) + (\text{sumofcyclingcodes} * W_{\text{cc}}) + (\text{sumofhealthcheckalerts} * W_{\text{h1}}) + (\text{sumofhealthcheckalarms} * W_{\text{h2}})])$$

[0149] As can be seen from the above example of FIG. **14**, time weighting (time varying weights) is considered in the performance checks. For example, the two warning codes **1440** and **1462** have values of 0.606 and 0.367, respectively, taking the time varying weights as described in relation to FIG. **12** into consideration. In the above example, the maximum value $I_{\text{sub},\max}=10$, and the overall performance index is constrained between 0 and $I_{\text{sub},\max}$, that is, between 0 and 10 in this example. The $I_{\text{sub},\max}$ can have other values. For example, if $I_{\text{sub},\max}=100$, the overall performance index is constrained between 0 and 100.

[0150] The following is an implementation of the generation of the performance index in pseudocode, in some embodiments. [0151] Read chiller time series data [0152] Read chiller codes: safety codes, warning codes, cycling codes, and operating codes [0153] Define inclusion interval=5 days [0154] Define reporting frequency=1 days [0155] Define weights={‘SAF-CODE’: 3.0, ‘WAR-CODE’: 1.0, ‘CYC-CODE’: 1.0, ‘health_check_alert’: 1.0, ‘health_check_alarm’: 2.0} [0156] Define range of index={‘min’: 0, ‘max’: 10} [0157] Define parameters using as threshold values for health check rules: [0158] param_values={‘percent_fl_min’: 10.0, [0159] ‘cond_app_alert’: 3.5, ‘cond_app_alarm’: 5, ‘cond_app_runtime’: 0.3, [0160] ‘low_ref_level_alert’: 15, ‘low_ref_level_alarm’: 10, ‘ref_level_runtime’: 0.5, [0161] ‘cond_entering_water_alarm’: 85, ‘cond_entering_water_runtime’: 0.05, [0162] ‘evap_app_alert’: 3.5, ‘evap_app_alarm’: 5, ‘evap_app_runtime’: 0.3, [0163] ‘high_oil_temp_alert’: 155, ‘high_oil_temp_alarm’: 165, ‘high_oil_temp_runtime’: 0.05, [0164] ‘high_ref_level_alarm’: 90, [0165] ‘low_cond_water_entering_temp_alert’: 49, ‘low_cond_water_entering_temp_alarm’: 47, [0166] ‘evap_delta_temp’: 1.5} [0167] For each day in dataset, skipping by reporting frequency: [0168] Look at window of (current date-inclusion interval) to current date; [0169] Identify any safety codes, warning codes, and cycle codes occurring in this window; [0170] Run health checks for all points in window (remove points when chiller wasn't running); [0171] Aggregate any violations of health checks and codes to produce index.

[0172] In some embodiments, the time varying performance indication system **502** can be configured to determine that a total runtime of the connect equipment in a past time window, and generate the overall performance index only when the total runtime of the connect equipment in the past time window satisfies a predetermined threshold. For example, the performance index may be calculated only for connected equipment that is considered to be running for more than a predetermined total runtime or threshold (e.g., 2 hours) in a time window (e.g., last 24-hour period). In some embodiments, if a chiller (or other connected equipment) is not considered to be running, the performance index may not show up in a heat map or metrics, but may be listed in a grey section that contains all chillers that are either not running or that have been flagged or marked as disregard. In some embodiments, the motor full-load amps (MOT FLA) point is being used to calculate whether or not a chiller is running for a health check. In other embodiments, a method that utilizes operational codes and potentially input power (INPUT KW) may be used.

[0173] In some embodiments, health checks can consider a predetermined rolling window (e.g., 30-day rolling window) for their frequency index calculations (time in alert or alarm divided by total runtime hours). In some embodiments, for the performance index, a different predetermined rolling window (e.g., 24-hour rolling window) may be used, but it may follow the same logic as far as putting in Alert or Alarm stage if the frequency index value is above 20%. In some embodiments, an example health check frequency index can be calculated as:

$$[00004]\text{HealthCheckFrequencyIndex} = \frac{\text{TimeinAlertorAlarminPast30Days}}{\text{TotalRuntimeinPast30Days}}$$

In some embodiments, an example performance frequency index can be calculated as:

$$[00005]\text{PerformanceFrequencyIndex} = \frac{\text{TimeinAlertorAlarminPast24Hours}}{\text{TotalRuntimeinPast24Hours}}$$

It should be understood that the example health check frequency index and/or the performance frequency index should be provided for illustrative purposes only and should not be regarded as limiting in any way. In some embodiments, the health check frequency index and/or the performance frequency index can be updated periodically (e.g., every 4 hours).

[0174] In some embodiments, the generated performance index can be rated and/or color coded to indicate if the overall health of the connected equipment should be concerned and further investigation should be conducted. For example, the following example threshold table, Table 4, can be used when the $I_{sub,max}=100$.

TABLE-US-00004 TABLE 4 threshold table for index values

Index Value Range	Rating	Color
75.0-100.0	Acceptable	Green
50.0-75.0	Alert	Yellow
0.0-50.0	Alarm	Red

[0175] In some embodiments, the time varying performance indication system **502** can cause an adjustment to the connect equipment based on the overall performance index generated for the connected equipment. For example, in some embodiments, a report can be generated to indicate the overall health of the connected equipment. The report can be transmitted via the network **446** to another system or device to prompt further investigation of the connected equipment. One or more adjustments or actions may be performed based on the further investigation as a result of the value of overall performance index.

[0176] Referring now to FIG. 15, an example user interface **1500** showing an example of the calculated performance index over time is shown, according to some embodiments. In some embodiments, the user interface **1500** can allow a user to hover over points and see any failed status checks and health checks that occurred during those times. In some embodiments, the user interface may also allow for the customization and tuning of parameters used in the algorithm, including how often the index is calculated, the date range to include, and weights for individual performance checks. In some embodiments, the user interface may provide an option to weigh the impact of failed performance checks by when they occurred. Such time weighting is beneficial because while users consider what has happened in the recent past, they may also want the index to be responsive to maintenance actions or other sudden changes in condition. In some embodiments, using exponentially or linearly decaying weights can enable the inclusion window adapt quickly to positive and negative changes in the index.

[0177] Referring now to FIG. 16, a flow diagram illustrating a process **1600** of generating a performance index for connected equipment is shown, according to some embodiments. The process **1600** can be performed by the time varying performance indication system **502** to automatically generating a performance index for connected equipment. For example, the processor **814** of the processing circuit **812** can be configured to perform the process **1600**. The process can include obtaining data points of a plurality of monitored variables and a plurality of status codes from past N time units (Stage **1602**). N is a number, such as an integer. The plurality of monitored variables are measured by connected equipment, and the plurality of status codes are generated by the connected equipment. In some embodiments, the time units can be days, hours, minutes, seconds, weeks, months, or years. The process can include obtaining a plurality of connected equipment specific parameters that are parameters specific to the connected equipment (Stage **1604**). The process can include performing a plurality of first performance checks for the connected equipment using the plurality of status codes from the past N time units (Stage **1606**). The process can include performing a plurality of second performance checks for the connected equipment using the data points of the plurality of monitored variables from the past N time units, the plurality of connected equipment specific parameters, and a plurality of predetermined rules (Stage **1608**). The process can include determining a plurality of individual performance check indicators based on the first performance checks and the second performance checks using a plurality of first weights each determined based on a different timing (Stage **1610**). The process can include generating an overall performance index for the connected equipment using the plurality of individual performance check indicators and a plurality of second weights (Stage **1612**). The process can include causing an adjustment to the connect equipment based on the overall performance index generated for the connected equipment (Stage **1614**).

[0178] Systems and methods described herein can aggregate a plurality of performance checks and generate an overall performance index as a metric for the overall health of a chiller or other connected equipment that can be tracked over time. By combining the health checks, cycling, warning, and safety codes, an aggregate health index for a chiller or other connected equipment can be calculated. This index can indicate the overall health of the connected equipment at a given time and provide field technicians and diagnostic engineers the ability to pinpoint which connected equipment is most critical to attend to. Tracking this performance index over time can draw attention to connected equipment that are consistently in poor health or that are trending in a negative direction, prompting branch service technicians to investigate. Connected equipment performance indices could also be aggregated across different machines to obtain an overall picture of the health of a specific customer's connected equipment or of the connected equipment serviced by a specific branch or region, providing better diagnosis abilities and results.

Configuration of Exemplary Embodiments

[0179] The construction and arrangement of the systems and methods as shown in the various exemplary embodiments are illustrative only. Although only a few embodiments have been described in detail in this disclosure, many modifications are possible (e.g., variations in sizes, dimensions, structures, shapes and proportions of the various elements, values of parameters, mounting arrangements, use of materials, colors, orientations, etc.). For example, the position of elements can be reversed or otherwise varied and the nature or number of discrete elements or positions can be altered or varied. Accordingly, all such modifications are intended to be included within the scope of the present disclosure. The order or sequence of any process or method steps can be varied or re-sequenced according to alternative embodiments. Other substitutions, modifications, changes, and omissions can be made in the design, operating conditions and arrangement of the exemplary embodiments without departing from the scope of the present disclosure.

[0180] The present disclosure contemplates methods, systems and program products on any machine-readable media for accomplishing various operations. The embodiments of the present disclosure can be implemented using existing computer processors, or by a special purpose computer processor for an appropriate system, incorporated for this or another purpose, or by a hardwired system. Embodiments within the scope of the present disclosure include program products comprising machine-readable media for carrying or having machine-executable instructions or data structures stored thereon. Such machine-readable media can be any available media that can be accessed by a general purpose or special purpose computer or other machine with a processor. By way of example, such machine-readable media can comprise RAM, ROM, EPROM, EEPROM, CD-ROM or other optical disk storage, magnetic disk storage or other magnetic storage devices, or any other medium which can be used to carry or store desired program code in the form of machine-executable instructions or data structures and which can be accessed by a general purpose or special purpose computer or other machine with a processor. Combinations of the above are also included within the scope of machine-readable media. Machine-executable instructions include, for example, instructions and data which cause a general purpose computer, special purpose computer, or special purpose processing machines to perform a certain function or group of functions.

[0181] Although the figures show a specific order of method steps, the order of the steps may differ from what is depicted. Also two or more steps can be performed concurrently or with partial concurrence. Such variation will depend on the software and hardware systems chosen and on designer choice. All such variations are within the scope of the disclosure. Likewise, software implementations could be accomplished with standard programming techniques with rule based logic and other logic to accomplish the various connection steps, processing steps, comparison steps and decision steps.

Claims

1. A system comprising: equipment operable to affect, monitor, or control one or more variable states or conditions in a building; a sensor configured to measure a variable relating to performance of the equipment; and a control system configured to: perform a plurality of performance checks for the equipment using a measurement from the sensor; determine a plurality of individual performance check indicators based on the plurality of

performance checks, the plurality of individual performance check indicators indicating events having different timings and severities; generate an overall performance index for the equipment using the plurality of individual performance check indicators and a plurality of weights based on the different timings and severities of the events indicated by the plurality of individual performance check indicators; and cause an adjustment to the equipment by initiating or executing a preventative maintenance action for the equipment based on the overall performance index.

2. The system of claim 1, wherein performing the plurality of performance checks comprises detecting deviations between a plurality of monitored variables associated with the equipment and a plurality of predetermined values.

3. The system of claim 1, wherein the plurality of weights comprise first weights based on the different timings and second weights based on the severities.

4. The system of claim 1, wherein generating the overall performance index for the equipment using the plurality of individual performance check indicators and the plurality of weights comprises: determining the timings of the events indicated by the plurality of individual performance check indicators; and using higher weights for events that have more recent timings and lower weights for events that have less recent timings.

5. The system of claim 1, wherein generating the overall performance index for the equipment using the plurality of individual performance check indicators and the plurality of weights comprises: determining the severities of the events indicated by the plurality of individual performance check indicators; and using higher weights for events that have higher severities and lower weights for events that have lower severities.

6. The system of claim 1, wherein: the events comprise occurrences of a plurality of alarms having different severities; and generating the overall performance index for the equipment comprises using a weight associated with an active alarm of the plurality of alarms having a highest severity of the different severities.

7. The system of claim 1, wherein the equipment comprises one or more chillers, air handling units, actuators, controllers, boilers, economizers, valves, dampers, cooling towers, fans, pumps, lighting equipment, security equipment, refrigeration equipment, HVAC equipment, or sensors.

8. The system of claim 1, wherein initiating or executing the preventative maintenance action comprises at least one of scheduling the preventative maintenance action in a maintenance scheduling system, providing an alert message to a user or a user interface, proactively repairing or replacing the equipment before a fault occurs, or adjusting a control algorithm for the equipment to repair the fault or work-around the fault.

9. A method comprising: operating equipment to affect, monitor, or control one or more variable states or conditions in a building; measuring, by a sensor, a variable relating to performance of the equipment; performing a plurality of performance checks for the equipment using a measurement from the sensor; determining a plurality of individual performance check indicators based on the plurality of performance checks, the plurality of individual performance check indicators indicating events having different timings and severities; generating an overall performance index for the equipment using the plurality of individual performance check indicators and a plurality of weights based on the different timings and severities of the events indicated by the plurality of individual performance check indicators; and causing an adjustment to the equipment by initiating or executing a preventative maintenance action for the equipment based on the overall performance index.

10. The method of claim 9, wherein performing the plurality of performance checks comprises detecting deviations between a plurality of monitored variables associated with the equipment and a plurality of predetermined values.

11. The method of claim 9, wherein the plurality of weights comprise first weights based on the different timings and second weights based on the severities.

12. The method of claim 9, wherein generating the overall performance index for the equipment using the plurality of individual performance check indicators and the plurality of weights comprises: determining the timings of the events indicated by the plurality of individual performance check indicators; and using higher weights for events that have more recent timings and lower weights for events that have less recent timings.

13. The method of claim 9, wherein generating the overall performance index for the equipment using the plurality of individual performance check indicators and the plurality of weights comprises: determining the severities of the events indicated by the plurality of individual performance check indicators; and using higher weights for events that have higher severities and lower weights for events that have lower severities.

14. The method of claim 9, wherein: the events comprise occurrences of a plurality of alarms having different severities; and generating the overall performance index for the equipment comprises using a weight associated with an active alarm of the plurality of alarms having a highest severity of the different severities.

15. The method of claim 9, wherein the equipment comprises one or more chillers, air handling units, actuators, controllers, boilers, economizers, valves, dampers, cooling towers, fans, pumps, lighting equipment, security equipment, refrigeration equipment, HVAC equipment, or sensors.

16. The method of claim 9, wherein initiating or executing the preventative maintenance action comprises at least one of scheduling the preventative maintenance action in a maintenance scheduling system, providing an alert message to a user or a user interface, proactively repairing or replacing the equipment before a fault occurs, or adjusting a control algorithm for the equipment to repair the fault or work-around the fault.

17. One or more non-transitory computer-readable media storing computer-executable instructions that, when executed by at least one processor, cause the at least one processor to perform operations comprising: operating equipment to affect, monitor, or control one or more variable states or conditions in a building; receiving a measurement from a sensor configured measure a variable relating to performance of the equipment; performing a plurality of performance checks for the equipment using the measurement; determining a plurality of individual performance check indicators based on the plurality of performance checks, the plurality of individual performance check indicators indicating events having different timings and severities; generating an overall performance index for the equipment using the plurality of individual performance check indicators and a plurality of weights based on the different timings and severities of the events indicated by the plurality of individual performance check indicators; and causing an adjustment to the equipment by initiating or executing a preventative maintenance action for the equipment based on the overall performance index.

18. The non-transitory computer-readable media of claim 17, wherein performing the plurality of performance checks comprises detecting deviations between a plurality of monitored variables associated with the equipment and a plurality of predetermined values.

19. The non-transitory computer-readable media of claim 17, wherein generating the overall performance index for the equipment using the plurality of individual performance check indicators and the plurality of weights comprises: determining the timings of the events indicated by the plurality of individual performance check indicators; and using higher weights for events that have more recent timings and lower weights for events that have less recent timings.

20. The non-transitory computer-readable media of claim 17, wherein generating the overall performance index for the equipment using the plurality of individual performance check indicators and the plurality of weights comprises: determining the severities of the events indicated by the plurality of individual performance check indicators; and using higher weights for events that have higher severities and lower weights for events that have lower severities.
